Michigan Tech

Definitions

Course reference

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Math Sciences

MA4535 — Nonlinear Dynamics & Chaos | Definitions | September 27, 2026

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Differential Equations
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Definition (Differential Equation)

An equation relating an unknown function to one or more of its derivatives.

Definition (Ordinary Differential Equation (ODE))

A differential equation whose unknown function depends on one independent variable, usually time t.

Definition (ODE System)

Coupled ordinary differential equations for the components of a state $\boldsymbol{y}(t)$. A first-order system has the form $\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y})$.

Initial Value Problem

Definition (Initial Value Problem (IVP))

A differential equation together with the value of the unknown state at a specified initial time.

For a second-order equation, the initial data usually include position and velocity:

$$y'' = F(t, y, y'), \qquad y(t_0) = y_0, \qquad y'(t_0) = v_0.$$

Flow, Trajectory, and Orbit

Poincare Sections and Return Maps

For an autonomous system $\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y})$ with unique solutions:

Definition (Flow)

The family of time maps φ_t sending an initial state $\boldsymbol{y}_0$ to its state at time t, wherever the solution exists.

Definition (Trajectory)

The time-dependent solution $t \mapsto \varphi_t(\boldsymbol{y}_0)$.

Definition (Orbit)

The set of states traced by a trajectory.

Stable and unstable equilibria

Let $\boldsymbol{q}^*$ be an equilibrium of $\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y})$, and let $\boldsymbol{y}(0) = \boldsymbol{y}_0$.

Definition (Equilibrium Point)

A state $\boldsymbol{q}^*$ satisfying $\boldsymbol{f}(\boldsymbol{q}^*) = \boldsymbol{0}$. Its trajectory is constant.

Definition (Stable equilibrium)

For every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$\|\boldsymbol{y}_0 - \boldsymbol{q}^*\| < \delta \quad \Longrightarrow \quad \|\boldsymbol{y}(t) - \boldsymbol{q}^*\| < \varepsilon \quad \text{for all } t \geq 0.$$

Definition (Unstable equilibrium)

An equilibrium that is not stable: some trajectories starting arbitrarily close leave a fixed neighborhood at a later time.

Poincaré Sections and Crossings

Sections and Return Maps

Definition (Poincaré Section)

A smooth hypersurface used to record transverse trajectory crossings. For $\Sigma = \{\boldsymbol{y} : h(\boldsymbol{y}) = 0\}$, require $\nabla h \neq \boldsymbol{0}$ on Σ.

Definition (Transverse Crossing)

A crossing at $\boldsymbol{q}$ where $\nabla h(\boldsymbol{q}) \cdot \boldsymbol{f}(\boldsymbol{q}) \neq 0$. The subsets Σ^+ and Σ^- have positive and negative values of this dot product, respectively. Tangencies belong to neither.

Definition (Crossing Points)

A recorded state $\boldsymbol{q}_j = \boldsymbol{y}(t_j)$ on one chosen crossing subset, with $t_{j+1} > t_j$.

First Return Time and Map
Sections and Return Maps

Definition (First Return Time)

The first positive time $\tau(\boldsymbol{q})$ at which the trajectory from $\boldsymbol{q}$ returns to the same chosen crossing subset. Set $\tau(\boldsymbol{q}) = +\infty$ if it never returns.

Definition (First Return Map)

The map $\boldsymbol{p}(\boldsymbol{q}) = \varphi_{\tau(\boldsymbol{q})}(\boldsymbol{q})$ on the subset of starting points with finite return time.

Periodic Solutions and Orbits
Periodic Orbits and Winding Numbers

Definition (Periodic Solution)

A solution satisfying $\boldsymbol{y}(t+T) = \boldsymbol{y}(t)$ for some $T > 0$ and every t in its complete trajectory.

Definition (Minimal Period)

The smallest positive time T for which a nonconstant solution satisfies $\boldsymbol{y}(t+T) = \boldsymbol{y}(t)$ for every t. An equilibrium has no minimal period.

Definition (Periodic Orbit)

The closed path traced by a nonconstant periodic solution.

Fixed Points and Minimal Map Periods

Periodic Orbits and Winding Numbers

Definition (Fixed Point)

A point $\boldsymbol{q}^*$ with $\boldsymbol{p}(\boldsymbol{q}^*) = \boldsymbol{q}^*$.

Definition (Periodic Point and Minimal Map Period)

A point has minimal map period m if $\boldsymbol{p}^m(\boldsymbol{q}) = \boldsymbol{q}$ and $\boldsymbol{p}^j(\boldsymbol{q}) \neq \boldsymbol{q}$ for $1 \leq j < m$. The integer m counts returns, not elapsed time.

Multipliers and Period Doubling
Periodic Orbits and Winding Numbers

Definition (Multiplier)

An eigenvalue of the derivative $D(\boldsymbol{p}^m)(\boldsymbol{q})$ at a periodic point with minimal map period m.

Definition (Period-Doubling Bifurcation)

A bifurcation in which an orbit of the return map with minimal map period m changes stability and a nearby orbit with minimal map period $2m$ emerges. In the generic case, a simple multiplier crosses -1, with the required nondegeneracy conditions satisfied.

Manifolds and Surface Coordinates

Manifolds, Coordinates, and Derivatives

Definition (Smooth Manifold)

A space described near each point by k independent local coordinates, with smooth changes between overlapping coordinate descriptions.

Definition (Surface Parameterization)

A smooth local map $\psi : U \subset \mathbb{R}^2 \to \Sigma$ describing a surface patch, with independent derivative columns and a local inverse giving coordinates.

Hyperbolicity and Invariant Manifolds

Manifolds, Coordinates, and Derivatives

Definition (Hyperbolic Equilibrium)

An equilibrium $\boldsymbol{q}^*$ for which every eigenvalue of $D\boldsymbol{f}(\boldsymbol{q}^*)$ has nonzero real part.

Definition (Stable Manifold)

For a hyperbolic equilibrium, the states whose trajectories exist for all forward times and approach $\boldsymbol{q}^*$ as $t \to +\infty$.

Definition (Unstable Manifold)

For a hyperbolic equilibrium, the states whose trajectories exist for all backward times and approach $\boldsymbol{q}^*$ as $t \to -\infty$.

Tangent Vectors and Tangent Spaces
Manifolds, Coordinates, and Derivatives

Definition (Coordinate Tangent Vectors)

The vectors $\partial\psi/\partial u$ and $\partial\psi/\partial v$. They span the surface tangent plane and need not be normalized.

Definition (Tangent Space)

The vector space of velocities of smooth curves on the manifold through a given point.

Flow and Return-Map Derivatives
Manifolds, Coordinates, and Derivatives

Definition (Flow Sensitivity Matrix)

The derivative $D_{\boldsymbol{y}_0}\varphi_t(\boldsymbol{y}_0)$ measuring the first-order effect of initial-state changes at fixed time.

Definition (Return-Map Derivative)

The derivative $D\boldsymbol{p}(\boldsymbol{q})$ measuring first-order changes in the return point for perturbations tangent to the section. It includes the effect of the changing return time.

References

[1] Yanghong Huang.
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